Advanced Coordinate Geometry

ANSWER KEY

Distance, midpoint, and slope combined

1 Triangle ABC has vertices $A(-2, 1)$, $B(4, 1)$, and $C(1, 7)$. Find the length of the median from C to the midpoint of AB.

Midpoint of AB: $\left(\frac{-2+4}{2}, \frac{1+1}{2}\right) = (1, 1)$. Length of median from $C(1, 7)$ to $(1, 1)$: $\sqrt{(1-1)^2 + (7-1)^2} = \sqrt{0+36} = 6$.

Equations of circles from given conditions

2 Find the equation of a circle with a diameter having endpoints $(2, 3)$ and $(8, 11)$.

Center (midpoint): $\left(\frac{2+8}{2}, \frac{3+11}{2}\right) = (5, 7)$. Radius (half the diameter length): diameter $= \sqrt{(8-2)^2 + (11-3)^2} = \sqrt{36+64} = \sqrt{100} = 10$, so radius $= 5$. Equation: $(x-5)^2 + (y-7)^2 = 25$.

Distance from a point to a line

3 Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

$d = \frac{|3(3) - 4(4) + 5|}{\sqrt{9+16}} = \frac{|9-16+5|}{5} = \frac{|-2|}{5} = \frac{2}{5}$.

Locus problems

4 Find the equation of the locus of points equidistant from $(2, 0)$ and $(-2, 0)$.

The locus of points equidistant from two fixed points is the perpendicular bisector of the segment joining them. Since both points lie on the x-axis symmetric about the origin, the perpendicular bisector is the vertical line $x = 0$ (the y-axis).

Rotations and general transformations

5 A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \to (-y, x)$.

Applying the rule: $(4, 0) \to (-0, 4) = (0, 4)$.

Dilations and scale factor

6 A triangle with vertices $(1, 2)$, $(3, 2)$, $(1, 6)$ is dilated by a scale factor of 3 centered at the origin. Find the coordinates of the image vertices.

Multiply each coordinate by 3: $(3, 6)$, $(9, 6)$, $(3, 18)$.

Finding the equation of a parabola from three points

7 A parabola $y = ax^2 + bx + c$ passes through $(0, 3)$, $(1, 6)$, and $(-1, 2)$. Find a, b, and c.

From $(0, 3)$: $c = 3$. From $(1, 6)$: $a + b + c = 6$, so $a + b = 3$. From $(-1, 2)$: $a - b + c = 2$, so $a - b = -1$. Adding $a + b = 3$ and $a - b = -1$: $2a = 2$, so $a = 1$, then $b = 2$. Equation: $y = x^2 + 2x + 3$.

A multi-part triangle-in-the-plane problem

8 This question has two parts. Triangle PQR has $P(0, 0)$, $Q(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula $\text{Area} = \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.

(a) Area $= \frac{1}{2}|0(0 - 6) + 6(6 - 0) + 3(0 - 0)| = \frac{1}{2}|0 + 36 + 0| = 18$. (b) Using base $PQ = 6$ (along the x-axis) and height equal to R's y-coordinate, 6: Area $= \frac{1}{2}(6)(6) = 18$ ✓, confirming both methods agree.

Finding the equation of a line given a point and a perpendicular condition

9 Find the equation of the perpendicular bisector of the segment connecting $(2, 3)$ and $(8, 7)$.

Midpoint: $\left(\frac{2 + 8}{2}, \frac{3 + 7}{2}\right) = (5, 5)$. Slope of the segment: $\frac{7 - 3}{8 - 2} = \frac{4}{6} = \frac{2}{3}$. Perpendicular slope: $-\frac{3}{2}$. Equation: $y - 5 = -\frac{3}{2}(x - 5)$, so $y = -\frac{3}{2}x + \frac{15}{2} + 5 = -\frac{3}{2}x + \frac{25}{2}$.

Translating and reflecting equations of curves

10 The graph of $y = x^2$ is reflected over the x-axis, then shifted 3 units up. Write the equation of the resulting graph.

Reflecting over the x-axis: $y = -x^2$. Shifting up 3: $y = -x^2 + 3$.

Finding vertices of a polygon given transformations

11 A square has vertices $(0, 0)$, $(4, 0)$, $(4, 4)$, $(0, 4)$. Find the coordinates after translating the square 3 units right and 2 units down.

Add $(3, -2)$ to each vertex: $(3, -2)$, $(7, -2)$, $(7, 2)$, $(3, 2)$.

Finding the intersection of a line and a parabola

12 Find the points of intersection of $y = x^2 - 2x - 3$ and $y = x - 3$.

Set equal: $x^2 - 2x - 3 = x - 3$, so $x^2 - 3x = 0$, giving $x(x - 3) = 0$, so $x = 0$ or $x = 3$. Corresponding y-values: for $x = 0$, $y = -3$; for $x = 3$, $y = 0$. Intersection points: $(0, -3)$ and $(3, 0)$.

A multi-part quadrilateral classification problem

13 This question has three parts. Quadrilateral $WXYZ$ has vertices $W(0, 0)$, $X(6, 0)$, $Y(8, 4)$, $Z(2, 4)$. (a) Find the length of each side. (b) Find the slopes of WX and ZY, and of WZ and XY. (c) Based on parts (a) and (b), classify the quadrilateral as precisely as possible.

(a) $WX = \sqrt{(6-0)^2 + 0^2} = 6$. $XY = \sqrt{(8-6)^2 + 4^2} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}$. $YZ = \sqrt{(2-8)^2 + 0^2} = 6$. $ZW = \sqrt{(0-2)^2 + (-4)^2} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}$. (b) Slope of WX: $\frac{0-0}{6-0} = 0$. Slope of ZY: $\frac{4-4}{8-2} = 0$ (parallel, since equal slopes). Slope of WZ: $\frac{4-0}{2-0} = 2$. Slope of XY: $\frac{4-0}{8-6} = 2$ (also parallel, equal slopes). (c) Since both pairs of opposite sides are parallel ($WX \parallel ZY$ and $WZ \parallel XY$) and opposite sides are equal in length ($WX = YZ = 6$, $XY = ZW = 2\sqrt{5}$), the quadrilateral is a parallelogram.

Finding the equation of a tangent line to a circle

14 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.

The radius to $(3, 4)$ has slope $\frac{4-0}{3-0} = \frac{4}{3}$. The tangent line is perpendicular, so its slope is $-\frac{3}{4}$. Equation: $y - 4 = -\frac{3}{4}(x - 3)$, so $y = -\frac{3}{4}x + \frac{9}{4} + 4 = -\frac{3}{4}x + \frac{25}{4}$.

Finding the vertex and axis of symmetry of a parabola from three points

15 A parabola passes through $(-1, 0)$, $(3, 0)$, and $(1, -8)$. Find the axis of symmetry and the vertex.

Since the parabola has zeros at $x = -1$ and $x = 3$, the axis of symmetry lies midway between them: $x = \frac{-1 + 3}{2} = 1$. Since $(1, -8)$ lies exactly on the axis of symmetry $x = 1$, this point IS the vertex: $(1, -8)$.